CIS 6270 Practice Exam 2 for Exam 2

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Use the following conditions for a timed practice attempt:

- Please write your name on each page. Allow yourself 90 minutes.
- Work through the exam on your own before consulting the solutions. Write down any questions you would like to discuss with the instructor or TAs.
- Each part has six choices. Use the space provided to work through the problem, then box one answer letter. Only the boxed letter counts toward your score.
- Each labeled part is worth 0.5 points. We award full credit for a correct answer and zero otherwise.
- You may use one side of an 8.5in $\times$ 11in cheat sheet containing only material from this class. Keep your signed sheet with your completed work. Do not use a calculator, phone, or other device during the timed attempt.

The University of Pennsylvania is a community dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. I will use this practice exam to assess my own understanding. I will complete the timed attempt using my own work before consulting the solutions or discussing answers.

This practice exam contains 12 points and does not contribute to your course grade.

Name: _____

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This practice exam is worth 12 points.

Fewer Sampling Steps with Class-Rate Flow Matching

You lead a group that designs peptide binders. An autoregressive model generates one residue per network call, so a 24-residue peptide takes 24 calls. With discrete flow matching, one call predicts residues at every position in parallel. To generate the same peptide in fewer calls, we would therefore need fewer than 24 sampling steps. For comparison, discrete flow matching samplers have used 1024 steps for text, 100 for images, and up to 10,000 for code.

What happens if we try to generate the entire peptide in one step? Each position is drawn independently given the same input, so a scaffold residue cannot depend on an anchor chosen in that step. In this exam, you will develop class-rate flow matching to reveal the anchors that carry the binding motif earlier on average. When we generate the remaining scaffold residues, we can then condition on the anchors already in place. During training, we still corrupt every position at randomly sampled times.

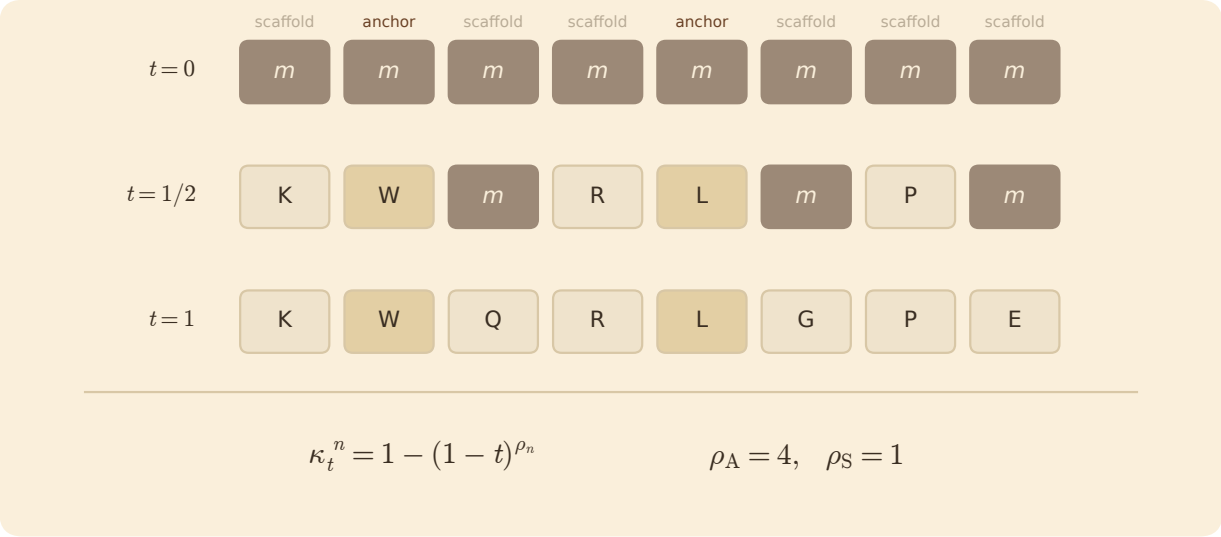


Figure 1. Class-rate flow matching on a peptide. We start with an all-mask peptide at $t = 0$ and generate a clean peptide at $t = 1$. A position of class n holds its data residue with probability $\kappa_t^n = 1 - (1 - t)^{\rho_n}$. We use a larger exponent ρ_n for anchor positions. The figure shows eight of the 24 positions. The row at $t = 1/2$ shows one sample from the path.

You will first use jump rates to describe how probability moves between peptide sequences. You will then choose a path for the anchor and scaffold residues, derive its velocity, and train a denoiser to sample from it. After examining how a corrector can revise a residue, you will consider a continuous flow on the simplex. Finally, you will guide generation toward binders with Doob's h-transform and interpret hypothetical results.

Research brief and notation

Each peptide $x = (x^1, \ldots, x^N)$ has $N = 24$ positions. Each position holds one of the 20 amino acids in $\mathcal{V}$, while $m \notin \mathcal{V}$ denotes the mask. We assign each position i to a class $c(i) \in \{\mathrm{A}, \mathrm{S}\}$. The six anchor positions carry the binding motif. The remaining 18 positions form the scaffold. In discrete flow matching, time runs from the source at $t = 0$ to the data at $t = 1$. We write the destination first, so $u_t(y, x)$ is the rate of jumping from x to y.

We start from an all-mask peptide and draw the clean peptide x_1 from the data distribution q. The probability of revealing a residue depends on its class,

$$\kappa_t^n = 1 - (1-t)^{\rho_n}, \qquad \rho_{\mathrm{A}} = 4, \qquad \rho_{\mathrm{S}} = 1,$$

so a position of class n holds its data residue with probability κ_t^n and remains masked with probability $1 - \kappa_t^n$. Setting $\rho_n = 1$ recovers the linear schedule $\kappa_t = t$. If we use this schedule for both classes, every position has the same reveal rate.

The denoiser $p_{1|t}^\theta(a \mid x)$ at position i is the probability that the data residue x_1^i is $a \in \mathcal{V}$, given the current peptide x. We write $x^{i \leftarrow a}$ for x with position i set to a, and δ for the indicator that its two arguments agree. Jumps change one position at a time, and $u_t^i(a, x)$ denotes the per-position velocity. For the parts on the simplex, we use the toy four-residue alphabet $\{\mathrm{L}, \mathrm{I}, \mathrm{K}, \mathrm{R}\}$. Logarithms are natural.

Step	**Points**
1. Read rates as velocities	2.0
2. Choose the class-rate path	2.0
3. Derive the velocity and the loss	2.0
4. Sample, correct, and check optimality	2.0
5. Relax the peptide to the simplex	2.0
6. Guide generation and read the results	2.0
Total: 24 parts	**12.0**

Step 1: Reading Rates as Velocities (2 points)

Before choosing a schedule for the anchors, we need to understand how jump rates change state probabilities. We can then check which dependencies we lose if we generate an entire peptide in one parallel step.

(a) We write each jump rate with the destination first. Which constraints must the rates satisfy for a fixed starting state?

A position currently holds the mask m, and its other states are the residues L, K, and R. Which vector $\big(u_t(\mathrm{L}, m),\ u_t(\mathrm{K}, m),\ u_t(\mathrm{R}, m),\ u_t(m, m)\big)$ is a valid probability velocity at this position? Choose ONE.

A. $(0.6,\ 0.9,\ 0.5,\ -1.0)$

B. $(0.6,\ -0.9,\ 0.5,\ -0.2)$

C. $(0.6,\ 0.9,\ 0.5,\ 2.0)$

D. $(0.2,\ 0.3,\ 0.5,\ 0)$

E. $(-0.6,\ -0.9,\ -0.5,\ 2.0)$

F. $(0.6,\ 0.9,\ 0.5,\ -2.0)$

(0.5 points)

(b) Once we know the jump rates, we can use them to calculate how the probability of each state changes.

At one position, the probabilities of $(\mathrm{L}, \mathrm{K}, \mathrm{R}, m)$ are (0.1, 0.2, 0.3, 0.4). From the mask, the rates of revealing L, K, and R are $u_t(\mathrm{L}, m) = 1$, $u_t(\mathrm{K}, m) = 2$, and $u_t(\mathrm{R}, m) = 1$, respectively. All other off-diagonal rates are zero. Which vector gives $\big(\dot{p}_t(\mathrm{L}),\ \dot{p}_t(\mathrm{K}),\ \dot{p}_t(\mathrm{R}),\ \dot{p}_t(m)\big)$? Choose ONE.

A. $(0.4,\ 0.8,\ 0.4,\ -1.6)$

B. $(1,\ 2,\ 1,\ -4)$

C. $(0.1,\ 0.4,\ 0.3,\ -0.8)$

D. $(0.4,\ 0.8,\ 0.4,\ -0.4)$

E. $(0.04,\ 0.16,\ 0.12,\ -0.32)$

F. $(-0.4,\ -0.8,\ -0.4,\ 1.6)$

(0.5 points)

(c) Suppose we sample two residues in parallel from a single network prediction. Can this step reproduce a dependence between the two positions?

A two-residue peptide forms a salt bridge, so its data distribution places probability 1/2 on LK and 1/2 on KL. From the all-mask peptide, we take one step that draws both positions at once from their own marginals. What is the probability that this step produces a data peptide? Choose ONE.

A. 1

B. 3/4

C. 1/2

D. 1/4

E. 1/8

F. 0

(0.5 points)

(d) To quantify the error in part (c), we compare the joint distribution with the product of its per-position marginals using KL divergence. Averaging this divergence over the source gives the conditional total correlation of the coupling.

For the data distribution in (c) with the all-mask source, what is the conditional total correlation? Choose ONE.

A. 0

B. $\frac{1}{2}\log 2$

C. $\log(4/3)$

D. $\log 2$

E. $\log 4$

F. $4\log 2$

(0.5 points)

Step 2: Choosing the Class-Rate Path (2 points)

With a shared schedule, anchor and scaffold positions have the same probability of remaining masked. We now give each class its own schedule so that anchors become visible earlier on average. To construct the velocity, we first need the reveal rate implied by each schedule.

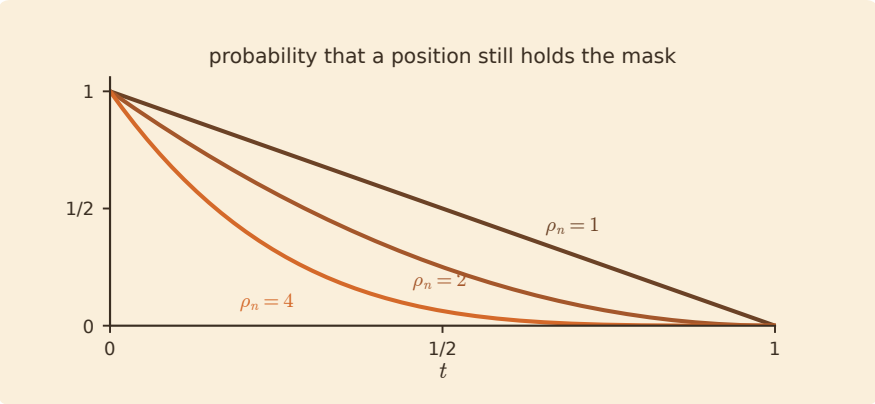


Figure 2. Probability that a position remains masked for three values of ρ_n. When $\rho_n = 1$, this probability decreases linearly with time.

(a) To find the reveal rate along a mixture path, we need the probability that a position is still masked and the rate at which that probability decreases.

With $\kappa_t^n = 1 - (1-t)^{\rho_n}$, what is the schedule factor $\dot{\kappa}_t^n/(1-\kappa_t^n)$? Choose ONE.

A. $\rho_n(1-t)^{\rho_n - 1}$

B. $\dfrac{\rho_n}{1-t}$

C. $\dfrac{1}{1-t}$

D. $\dfrac{\rho_n}{t}$

E. $\dfrac{\rho_n(1-t)^{\rho_n-1}}{1-(1-t)^{\rho_n}}$

F. $\dfrac{\rho_n}{(1-t)^{\rho_n}}$

(0.5 points)

(b) Now that we have the reveal rate, how does the probability of remaining masked change over time?

Let $p_t(m)$ be the probability that a position of class n still holds the mask. Which equation is the master equation at the mask state? Choose ONE.

A. $\dot{p}_t(m) = \dfrac{\rho_n}{1-t}\, p_t(m)$

B. $\dot{p}_t(m) = -\dfrac{\rho_n}{1-t}\big(1 - p_t(m)\big)$

C. $\dot{p}_t(m) = -\dfrac{\rho_n}{1-t}$

D. $\dot{p}_t(m) = -\dfrac{\dot{\kappa}_t^n}{\kappa_t^n}\, p_t(m)$

E. $\dot{p}_t(m) = \dfrac{\rho_n}{1-t}\big(1 - p_t(m)\big)$

F. $\dot{p}_t(m) = -\dfrac{\rho_n}{1-t}\, p_t(m)$

(0.5 points)

(c) To sample over a finite time interval in Step 4, we will need the probability that a position remains masked throughout that interval.

For $s < t$, what is the probability that a position of class n still holds the mask at time t, given that it holds the mask at time s? Choose ONE.

A. $\left(\dfrac{1-t}{1-s}\right)^{\rho_n}$

B. $(1-t)^{\rho_n}$

C. $\left(\dfrac{1-s}{1-t}\right)^{\rho_n}$

D. $\big(1-(t-s)\big)^{\rho_n}$

E. $\dfrac{(1-t)^{\rho_n}-(1-s)^{\rho_n}}{1-(1-s)^{\rho_n}}$

F. $1-\left(\dfrac{1-t}{1-s}\right)^{\rho_n}$

(0.5 points)

(d) Our main runs use $\rho_{\mathrm{A}} = 4$ at the anchors and $\rho_{\mathrm{S}} = 1$ at the scaffold.

At $t = 3/4$, which pair gives the probability that an anchor position still holds the mask and the probability that a scaffold position still holds the mask? Choose ONE.

A. $(1/16,\ 1/4)$

B. $(1/256,\ 1/4)$

C. $(1/64,\ 1/4)$

D. $(1/4,\ 1/256)$

E. $(255/256,\ 3/4)$

F. $(1/256,\ 3/4)$

(0.5 points)

Step 3: Deriving the Velocity and the Loss (2 points)

We have chosen how each position moves from the mask to its clean residue. We can now derive the jump rates that generate this path for a known clean peptide. During sampling, we only know the current peptide, so we will train a denoiser to predict the clean residues from it.

(a) When the clean peptide is known, the conditional velocity directs each masked position toward its clean residue. To obtain a velocity we can use during sampling, we average over possible clean residues using the probabilities predicted by the denoiser.

Write d for $p^{\theta}_{1|t}(a \mid x)$. Which pair gives the conditional velocity $u^i_t(a, z \mid x_1)$ from the current residue z at a position of class n, and the model velocity $u^i_t(a, x)$? Choose ONE.

A. $\left(\frac{\rho_n}{1-t}\,\delta(a, x^i_1),\quad \frac{\rho_n}{1-t}\,d\right)$

B. $\left(\frac{\rho_n}{1-t}\big[\delta(a, x^i_1) - \delta(a, z)\big],\quad \frac{\rho_n}{1-t}\big[\delta(a, x^i) - d\big]\right)$

C. $\left(\frac{\dot\kappa^n_t}{\kappa^n_t}\big[\delta(a, x^i_1) - \delta(a, z)\big],\quad \frac{\dot\kappa^n_t}{\kappa^n_t}\big[d - \delta(a, x^i)\big]\right)$

D. $\left(\frac{\rho_n}{1-t}\big[\delta(a, x^i_1) - \delta(a, z)\big],\quad \frac{\rho_n}{1-t}\big[d - \delta(a, x^i)\big]\right)$

E. $\left(\frac{1}{1-t}\big[\delta(a, x^i_1) - \delta(a, z)\big],\quad \frac{1}{1-t}\big[d - \delta(a, x^i)\big]\right)$

F. $\left(\rho_n(1-t)^{\rho_n - 1}\big[\delta(a, x^i_1) - \delta(a, z)\big],\quad \rho_n(1-t)^{\rho_n - 1}\big[d - \delta(a, x^i)\big]\right)$

(0.5 points)

(b) To train the denoiser, we sample a time t and a pair (x_0, x_1). We use the conditional path to draw a noisy peptide x_t, then evaluate the cross-entropy of the predicted residue distribution at each position.

Which expression is that loss, written for the class-rate path? Choose ONE.

A. $-\mathbb{E}\Big[\sum_{i=1}^{N} \frac{\rho_{c(i)}}{1-t} \log p^{\theta}_{1|t}(x_1^i \mid x_t)\Big]$

B. $-\mathbb{E}\Big[\sum_{i:\, x_t^i = m} \frac{\rho_{c(i)}}{1-t} \log p^{\theta}_{1|t}(x_1^i \mid x_t)\Big]$

C. $-\mathbb{E}\Big[\sum_{i=1}^{N} \frac{1}{1-t} \log p^{\theta}_{1|t}(x_1^i \mid x_t)\Big]$

D. $-\mathbb{E}\Big[\sum_{i:\, x_t^i = m} \log p^{\theta}_{1|t}(x_1^i \mid x_t)\Big]$

E. $-\mathbb{E}\Big[\sum_{i=1}^{N} \rho_{c(i)}(1-t)^{\rho_{c(i)}-1} \log p^{\theta}_{1|t}(x_1^i \mid x_t)\Big]$

F. $-\mathbb{E}\Big[\sum_{i=1}^{N} \log p^{\theta}_{1|t}(x_1^i \mid x_t)\Big]$

(0.5 points)

(c) We can score one training peptide by hand.

For a four-position peptide, the network assigns the correct residues probabilities 1/2, 1/4, 1/4, and 1/2, respectively. What is the total cross-entropy across the four positions? Choose ONE.

A. $\frac{3}{2}\log 2$

B. $2\log 2$

C. $3\log 2$

D. $4\log 2$

E. $6\log 2$

F. $12\log 2$

(0.5 points)

(d) Several clean peptides can produce the same noisy peptide. This uncertainty limits how low the denoising loss can fall.

At its minimum over θ, what does the expected loss in (b) equal? Choose ONE.

A. The expected entropy of the true denoiser $p_{1|t}(\cdot \mid x_t)$, summed over the positions.

B. Zero, because the network can match the data residue at every position.

C. The expected KL divergence from the true denoiser to the network's prediction.

D. The entropy of the data distribution q.

E. The expected variance of the conditional velocity.

F. The expected cross-entropy between the source distribution and the data distribution.

(0.5 points)

Step 4: Sampling, Correcting, and Checking Optimality (2 points)

Now that we have trained the denoiser, we can generate a peptide by starting from the all-mask source and taking Euler steps to $t = 1$. The schedule determines the probability of redrawing each position. We must choose the step length so that $\lambda_t \leq 1$.

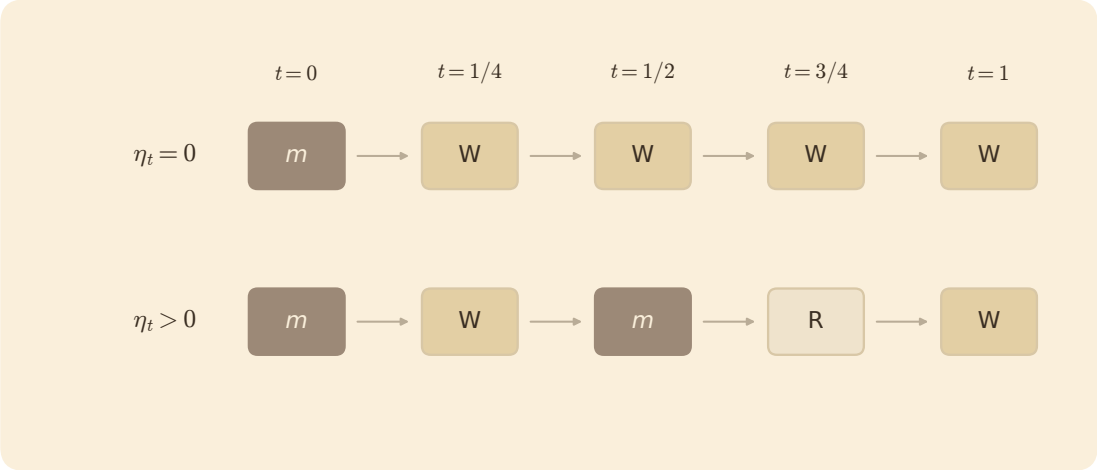


Figure 3. Two sample trajectories at one peptide position. Without a corrector, a revealed residue stays fixed. With a corrector, we can mask that position again and draw a new residue.

(a) At each step, we redraw a position from the denoiser with probability $\lambda_t = h\,\rho_n/(1-t)$ and keep its current residue otherwise.

With $h = 1/8$ at $t = 1/2$, which pair gives λ_t at an anchor position and at a scaffold position? Choose ONE.

A. $(1/2,\ 1/8)$

B. $(1,\ 1/4)$

C. $(1/4,\ 1)$

D. $(1/2,\ 1/2)$

E. $(2,\ 1/2)$

F. $(4,\ 1)$

(0.5 points)

(b) We take eight uniform steps from $t = 0$ to $t = 1$, so $h = 1/8$ and step k begins at $t = k/8$ for $k = 0, \ldots, 7$.

At which of these times does λ_t first rise strictly above one at an anchor position? Choose ONE.

A. $t = 1/8$

B. $t = 1/2$

C. $t = 5/8$

D. $t = 3/4$

E. $t = 7/8$

F. It stays at or below one at all eight times.

(0.5 points)

(c) To revise an early choice while preserving the probability path, we can add a corrector with zero net flux. The backward velocity at a position of class n is $\breve{u}_t^i(a, x) = \frac{\dot{\kappa}_t^n}{\kappa_t^n}\big[\delta(a, x^i) - p_{0|t}(a \mid x)\big]$. For the all-mask source, $p_{0|t}(\cdot \mid x) = \delta_m$.

Write d for $p_{1|t}^\theta(a \mid x)$, let $\hat{u}_t$ be the model velocity of 3(a), and take $\bar{u}_t = \hat{u}_t + \eta_t(\hat{u}_t - \breve{u}_t)$. Which pair gives the rate of revealing residue a at a masked position of class n and the rate of remasking a visible one? Choose ONE.

A. $\left(\frac{\rho_n}{1-t}\, d, \quad \eta_t \frac{\dot{\kappa}_t^n}{\kappa_t^n}\right)$

B. $\left((1+\eta_t)\frac{\rho_n}{1-t}\, d, \quad \eta_t \frac{\rho_n}{1-t}\right)$

C. $\left(\eta_t \frac{\rho_n}{1-t}\, d, \quad (1+\eta_t)\frac{\dot{\kappa}_t^n}{\kappa_t^n}\right)$

D. $\left((1+\eta_t)\frac{\rho_n}{1-t}\, d, \quad 0\right)$

E. $\left((1-\eta_t)\frac{\rho_n}{1-t}\, d, \quad \eta_t \frac{\dot{\kappa}_t^n}{\kappa_t^n}\right)$

F. $\left((1+\eta_t)\frac{\rho_n}{1-t}\, d, \quad \eta_t \frac{\dot{\kappa}_t^n}{\kappa_t^n}\right)$

(0.5 points)

(d) We can also choose a velocity by minimizing its kinetic energy along the conditional path. For the edge weight $w_t(y, x) = 1/p_t(y)$, the minimizing velocity is $u_t^\star(y, x \mid x_1) = p_t(y \mid x_1)\big[\partial_t \log p_t(y \mid x_1) - \partial_t \log p_t(x \mid x_1)\big]_+$. When the path depends on distances between residues, jumps occur only toward residues closer to the data residue.

Take the toy alphabet $\{\mathrm{L}, \mathrm{I}, \mathrm{K}, \mathrm{R}\}$ with $\Delta = 1$ inside the hydrophobic pair $\{\mathrm{L}, \mathrm{I}\}$ and inside the basic pair $\{\mathrm{K}, \mathrm{R}\}$, and $\Delta = 2$ across the pairs. The path is $p_t(y \mid x_1) \propto e^{-\beta_t \Delta(y, x_1)}$ with $\beta_t = t/(1-t)$. At $t = 1/2$ we have $\beta_t = 1$ and $\dot{\beta}_t = 4$, and with $x_1 = \mathrm{L}$ the path gives (0.6103, 0.2245, 0.0826, 0.0826) over $(\mathrm{L}, \mathrm{I}, \mathrm{K}, \mathrm{R})$. The current residue is K. Which rates does $u_t^\star(\cdot, \mathrm{K} \mid \mathrm{L})$ give toward $(\mathrm{L}, \mathrm{I}, \mathrm{R})$? Choose ONE.

A. (2.441, 0.449, 0)

B. (4.882, 0.898, 0)

C. (0, 0.898, 0.661)

D. (0.898, 4.882, 0)

E. (4.882, 1.796, 0)

F. (4.882, 0.898, 0.661)

(0.5 points)

Step 5: Relaxing the Peptide to the Simplex (2 points)

So far, each position has changed through discrete jumps. We can also represent a position by a probability vector and move it continuously toward a residue at a corner of the simplex. We then sample by integrating an ODE, which we can guide with a classifier or distill into a faster sampler. For the toy alphabet $\{\mathrm{L}, \mathrm{I}, \mathrm{K}, \mathrm{R}\}$, the simplex is a tetrahedron with corners $e_{\mathrm{L}}, e_{\mathrm{I}}, e_{\mathrm{K}}, e_{\mathrm{R}}$.

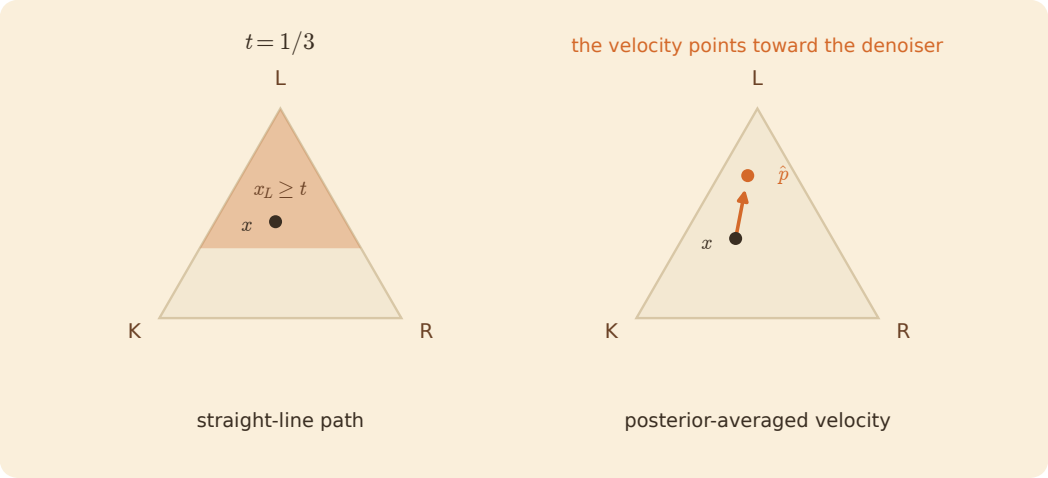


Figure 4. A triangle stands for the simplex over three of the four residues, with two separate points x. On the left, the points that could have come from the corner e_{L} under the straight-line path satisfy $x_{\mathrm{L}} \geq t$. On the right, the marginal velocity at a different point runs toward the denoiser $\hat{p}$.

(a) Consider the straight-line path $\psi_t = (1-t)x_0 + t\, e_k$ with x_0 drawn uniformly on the simplex. A point on a path ending at corner e_k must satisfy $x_k \geq t$. We can use this condition together with the fact that the coordinates sum to one.

At $t = 0.45$, at most how many of the four corners keep positive posterior probability at a position? Choose ONE.

A. 0

B. 1

C. 2

D. 3

E. It depends on the point x, so no bound holds at a fixed time.

F. All four, because every corner keeps positive probability at every finite time.

(0.5 points)

(b) The marginal velocity averages the conditional velocities over the corners, weighted by the denoiser.

At $t = 1/2$ a position sits at $x = (0.40, 0.30, 0.20, 0.10)$ and the denoiser gives $\hat{p} = (0.50, 0.20, 0.20, 0.10)$ over $(\mathrm{L}, \mathrm{I}, \mathrm{K}, \mathrm{R})$. With the conditional velocity $(e_k - x)/(1-t)$ toward corner e_k, what is the marginal velocity at this position? Choose ONE.

A. $(0.20,\ -0.20,\ 0,\ 0)$

B. $(0.10,\ -0.10,\ 0,\ 0)$

C. $(0.40,\ -0.40,\ 0,\ 0)$

D. $(1.00,\ 0.40,\ 0.40,\ 0.20)$

E. $(0.20,\ 0.20,\ 0,\ 0)$

F. $(-0.20,\ 0.20,\ 0,\ 0)$

(0.5 points)

(c) To measure Fisher-Rao distance on the simplex, we first apply the square-root map to each probability vector. We can then compute the distance from the angle between the resulting points on the unit sphere.

What is the Fisher-Rao distance from the uniform point (1/4, 1/4, 1/4, 1/4) to the corner e_{L}? Choose ONE.

A. $\pi/3$

B. $\pi/2$

C. $2\pi/3$

D. $3\pi/4$

E. $2\arccos(1/4)$

F. π

(0.5 points)

(d) The network can predict a velocity with a component normal to the sphere. To move along the sphere, we subtract that component.

At the sphere point $x = (1/2,\ 1/2,\ 1/2,\ 1/2)$ the network outputs $v = (0.4,\ 0.2,\ -0.2,\ 0.0)$. Which velocity do we use? Choose ONE.

A. $(0.40,\ 0.20,\ -0.20,\ 0.00)$

B. $(0.30,\ 0.10,\ -0.30,\ -0.10)$

C. $(0.50,\ 0.30,\ -0.10,\ 0.10)$

D. $(0.20,\ 0.00,\ -0.40,\ -0.20)$

E. $(0.10,\ -0.10,\ -0.50,\ -0.30)$

F. $(0.30,\ 0.10,\ -0.30,\ 0.10)$

(0.5 points)

Step 6: Guiding Generation and Reading the Results (2 points)

So far, we have generated peptides without controlling whether they bind the target. To guide generation toward binders, we can keep the trained denoiser and condition on a binding event $\mathcal{B}$ using Doob's h-transform. Here $\mathcal{B}$ is the event that a binding classifier accepts the finished peptide. Given the current peptide, the probability of this event is $h_t(x) = \Pr(\mathcal{B} \mid x_t = x)$.

(a) The value $h_t(x)$ is the probability of the event at the end of generation, starting from the current peptide.

Write d for $p^{\theta}_{1|t}(a \mid x)$. Which expression is the guided rate of revealing residue a at a masked position of class n? Choose ONE.

A. $\dfrac{\rho_n}{1-t}\, d\, \dfrac{h_t(x)}{h_t(x^{i \leftarrow a})}$

B. $\dfrac{\rho_n}{1-t}\, d\, h_t(x^{i \leftarrow a})$

C. $\dfrac{\rho_n}{1-t}\, d\, \dfrac{h_t(x^{i \leftarrow a})}{h_t(x)}$

D. $\dfrac{\dot{\kappa}^n_t}{\kappa^n_t}\, d\, \dfrac{h_t(x^{i \leftarrow a})}{h_t(x)}$

E. $\dfrac{\rho_n}{1-t}\, d + h_t(x^{i \leftarrow a}) - h_t(x)$

F. $\dfrac{\rho_n}{1-t}\, d\, \dfrac{h_t(x^{i \leftarrow a})}{1 - h_t(x)}$

(0.5 points)

(b) A classifier on noisy peptides can be hard to train, so instead, we can estimate the value by completing the peptide with the model and scoring each completion. To adjust the strength of guidance, we raise the value ratio to a power γ before multiplying it by the unguided rate.

At a masked anchor position at $t = 1/2$, the unguided rates toward L and K are 4 and 2. Five completions after revealing L earn rewards $(1, 1, 0, 1, 1)$, and five after revealing K earn $(0, 1, 0, 0, 0)$. The current value estimate is $1/2$, and we take $\gamma = 2$. Given that the next reveal at this position is L or K, what is the probability that it is L? Choose ONE.

A. 2/3

B. 4/5

C. 8/9

D. 16/17

E. 32/33

F. 64/65

(0.5 points)

Suppose we train class-rate flow matching on a set of peptide binders. We vary ρ_{A} while holding $\rho_{\mathrm{S}} = 1$ fixed. We will use the hypothetical results in Table 1 and Figure 5 to examine how the anchor schedule affects recovery at two sampling budgets.

Model	$\boldsymbol{\rho_{\mathrm{A}}}$	**8 steps**	**128 steps**
Autoregressive	–	–	61.4
Class-rate (uniform)	1	42.3	60.9
Class-rate	2	48.1	61.1
Class-rate	4	55.6	61.2
Class-rate	8	52.0	60.6
Class-rate	16	44.7	58.9

Table 1. Hypothetical anchor recovery (%) on held-out binders, at 8 and at 128 uniform sampling steps. The autoregressive model takes one network call per residue, so it costs 24 calls and has no eight-step entry.

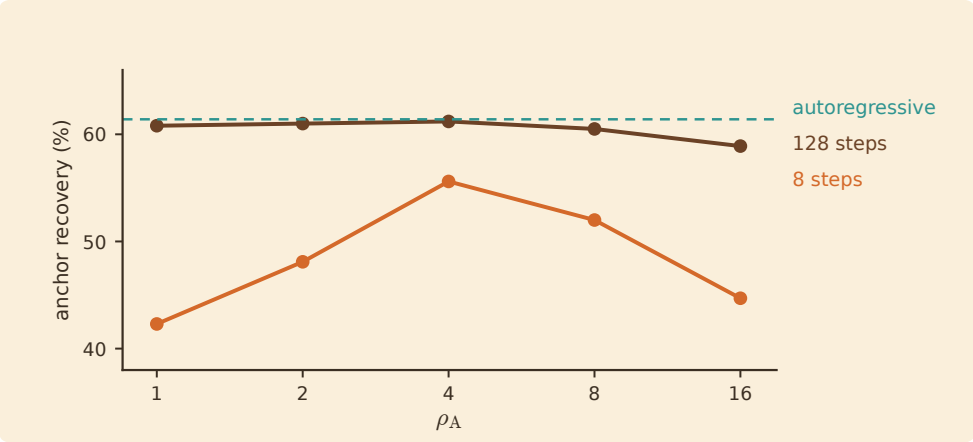


Figure 5. The values of Table 1 against ρ_A on a logarithmic axis. The dashed line marks the autoregressive model.

(c) With eight sampling steps, recovery rises from $\rho_A = 1$ to $\rho_A = 4$ before falling at larger exponents.

How can we explain this trend given the eight-step sampling budget? Choose ONE.

A. Larger ρ_A reveals the anchors earlier and helps at every value, so the fall from 55.6 to 44.7 after $\rho_A = 4$ reflects noise in the evaluation.

B. Larger ρ_A lowers λ_t at the anchors, so fewer of the eight steps are clipped and recovery rises across the whole sweep.

C. Larger ρ_A raises the loss weight at the anchors, so the denoiser fits them at the expense of the scaffold.

D. Revealing anchors earlier helps us predict scaffold residues. Beyond $\rho_A = 4$, more of the eight steps have $\lambda_t > 1$, so the cost of clipping exceeds the benefit of revealing anchors early.

E. Larger ρ_A needs more network calls per step at the anchor positions, so the eight-step budget buys fewer reveals there and recovery falls once the anchors outpace it.

F. Recovery at eight steps must sit below recovery at 128 steps, so the eight-step column follows the 128-step column and nothing about ρ_A follows.

(0.5 points)

(d) With 128 sampling steps, anchor recovery stays near 61 percent for every ρ_{A} up to 8.

Which reading of that column agrees with the construction? Choose ONE.

A. The class rates cancel from the loss, so every value of ρ_{A} trains the same denoiser, and the 8-step and 128-step columns must therefore show the same values.

B. At 128 steps the redraw probability λ_{t_k} stays at or below one for every ρ_{A} in the table, so no step is clipped and each sampler follows its own path exactly.

C. The denoiser reaches zero loss at 128 steps, so recovery is limited by the data distribution alone.

D. The paths share a source at $t = 0$ and differ in where they end at $t = 1$, so the models agree early in generation and a long step budget hides the difference between their endpoints.

E. Every schedule has $\kappa_0^n = 0$ and $\kappa_1^n = 1$, so all paths end at the data distribution. With 128 steps, the sampling error is small.

F. Anchor recovery saturates at 61.4, so no model can be separated from the autoregressive one at any step count.

(0.5 points)

End of practice exam.

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